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Optimal Partitioning Changepoint Analysis

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Abstract

Detecting changepoints in time series is a fundamental task in statistical modeling and data-driven decision-making. We introduce a novel set covering-based model for changepoint detection that leverages combinatorial optimization to identify an optimal set of segments explaining the observed data. Unlike conventional methods based on dynamic programming (DP), which often impose strict structural constraints on the objective function (e.g., additivity), our formulation uses Mixed-Integer Linear Programming (MILP). This approach offers significantly increased expressiveness and flexibility in the cost structure, accommodating diverse, non-additive loss functions and complex penalty schemes. Our formulation guarantees global optimality under this flexible cost structure, a critical advantage over heuristic or approximate approaches. The model's design enables high adaptability to different application domains, including finance, bioinformatics, and industrial monitoring. The increased efficiency of standard MILP solvers, coupled with tailored dominance rules, ensures practical scalability, achieving computational efficiency on datasets with several thousands of observations. Extensive experiments demonstrate that our approach outperforms state-of-the-art methods in both accuracy and runtime, offering a robust and interpretable solution for optimal changepoint detection.

Keywords: Time series; Set covering; segmentation; change point detection

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1. Introduction

Changepoint detection (CPD) is the task of identifying abrupt changes in the statistical properties of a time series and constitutes a foundational problem across statistics, machine learning, and signal processing [1–3]. The objective of CPD is to partition a temporal sequence into contiguous, non-overlapping segments, each characterized by internally homogeneous behavior. Such segmentation enables a clearer understanding of the underlying data-generating process and facilitates the application of targeted analytical or predictive models at the segment level. As a result, effective changepoint detection plays a critical role not only in descriptive analysis, but also in forecasting, anomaly detection, and data-driven decision-making.

CPD plays a critical role in a wide range of application domains, including financial market analysis (regime shifts), industrial process monitoring (fault detection), climate science (trend changes), and bioinformatics (genomic variation analysis). Across these domains, a CPD method must satisfy two key requirements: (i) accurately segment the data so as to faithfully represent the underlying process, and (ii) provide a globally optimal solution, since suboptimal segmentation can propagate errors downstream and lead to significant real-world costs.

Time series segmentation has therefore attracted extensive attention, leading to the development of a broad spectrum of methodologies, including classical statistical techniques, Bayesian approaches, machine learning algorithms, and hybrid models. Among these, segmentation based on linear models remains particularly prominent due to its computational efficiency and interpretability. This setting, commonly referred to as piecewise linear regression [4], assumes that the time series can be approximated by a sequence of linear trends, each corresponding to a distinct segment.

For exact and globally optimal changepoint detection under additive cost structures, the dominant methodological paradigm is dynamic programming (DP). Methods such as Optimal Partitioning exploit the recursive structure of the problem, constructing the optimal segmentation for a sequence of length N from optimal solutions to shorter prefixes. This approach yields polynomial-time algorithms under suitable assumptions and has proven highly effective in practice. However, the DP framework relies fundamentally on the additivity of segment-wise costs, typically combined with a penalty proportional to the number of segments.

This additive and recursive structure imposes a significant modeling limitation. In many practical settings, the analyst wishes to enforce global structural constraints that couple properties of all selected segments. Examples include limiting the total duration of short segments, bounding the number of changepoints within specific time windows, or enforcing regularity in segment lengths to avoid excessively fragmented solutions. While certain such constraints can, in principle, be approximated through state augmentation or penalty tuning, doing so rapidly compromises the tractability and scalability of DP-based methods. As a result, DP frameworks remain largely confined to locally additive objectives, restricting their applicability in scenarios where global control over the segmentation structure is essential.

To overcome this limitation, we propose a novel formulation of the CPD problem as a combinatorial optimization model, specifically a set covering problem. In this formulation, all feasible segments are pre-enumerated and treated as decision variables, and the task of changepoint detection becomes the selection of a minimum-cost subset of segments that exactly covers the time series. Unlike breakpoint-indicator or network-flow formulations, this approach directly models segments as atomic objects, naturally enabling rich interactions among them.

The key advantage of the MILP framework lies not only in its ability to reproduce the classical additive cost structure, but more importantly in its capacity to enforce complex global constraints through simple linear inequalities. These constraints can link arbitrary subsets of segment variables, control aggregate properties such as the number, length, or distribution of segments, and encode domain-specific structural requirements that are difficult or impractical to express within DP-based models. Unlike breakpoint-indicator or network-flow formulations, the proposed set-covering approach provides a direct and flexible mechanism for global control over the segmentation. While MILP solving is NP-hard in general, recent advances in solver technology, together with problem-specific preprocessing and bounding techniques, make this approach computationally viable for problem sizes of practical interest.

In this paper, we introduce Set-Covering based Changepoint Detection (SC-CPD), a novel MILP framework for optimal time series segmentation. Our main contributions are as follows:

A set-covering MILP formulation for CPD with global structural control. We formulate the changepoint detection problem as a set covering model, enabling the direct incorporation of global constraints that are beyond the practical reach of dynamic programming methods.

Enhanced modeling fidelity through global constraints. We demonstrate the practical value of the proposed framework by introducing and analyzing several global constraints that improve the interpretability, robustness, and domain relevance of the resulting segmentations.

Scalability through preprocessing and dominance rules. We develop specialized preprocessing techniques, including dominance rules and bounding strategies, that significantly reduce the number of candidate segments and allow the model to scale to time series with several thousand observations.

Competitive computational performance and solution quality. Through extensive computational experiments, we show that SC-CPD produces higher-quality segmentations while achieving runtimes that are comparable to, and in many cases competitive with, state-of-the-art DP-based and heuristic methods.

The paper further illustrates the versatility of the proposed approach through a range of real-world applications, including financial time series, epidemiological data, and environmental monitoring. Common challenges such as noise, irregular sampling, and high-dimensional feature spaces are discussed, highlighting the advantages of MIP-based changepoint detection in addressing domain-specific structural requirements.

The remainder of this paper is organized as follows...

2. Related literature

The problem of optimal changepoint detection and time series segmentation has been extensively studied, and a wide range of exact and approximate methods have been proposed. Comprehensive surveys are provided in [2,3], which classify approaches into statistical, Bayesian, algorithmic, and learning-based families. In this section, we focus specifically on exact optimization-based formulations for changepoint detection and highlight their modeling assumptions and limitations relative to the proposed approach.

Dynamic programming (DP) constitutes the dominant paradigm for exact changepoint detection under additive cost structures. Classical methods such as Segment Neighborhood [5] and Optimal Partitioning [6] formulate the problem as the minimization of a sum of segment-wise losses plus a penalty on the number of changepoints. By exploiting the optimal substructure of the problem, these methods guarantee global optimality and admit polynomial-time algorithms under suitable assumptions.

Substantial effort has been devoted to improving the practical efficiency of DP-based CPD. Notable examples include the Pruned Exact Linear Time (PELT) algorithm [7], which uses inequality-based pruning to reduce the search space, and functional pruning approaches [8,9], which exploit convexity properties of the segment cost. These methods have become state of the art for large-scale problems with additive objectives.

Despite their efficiency, DP-based models fundamentally rely on the separability of the objective function across segments. This reliance severely limits their ability to incorporate constraints that couple decisions across the entire segmentation. While constrained DP and state-augmented formulations have been proposed for specific problems [10], the resulting state spaces typically grow exponentially, making such approaches impractical beyond very limited forms of global control. Consequently, DP methods remain best suited for problems with simple additive objectives and limited structural constraints.

Several optimal CPD models reinterpret the segmentation problem as a shortest-path problem on a directed acyclic graph, where nodes correspond to time indices and arcs represent feasible segments weighted by their associated costs [11,12]. This representation is then solved by DP models that yield efficient shortest-path algorithms.

Although graph-based formulations offer valuable conceptual insight and algorithmic flexibility, they inherit the same structural limitations as the DP at their core. In particular, it is difficult to express global constraints that do not decompose additively along a path without introducing additional dimensions or complex side constraints, which compromise tractability once again.

Mixed-Integer Programming (MIP) has been previously applied to changepoint detection and piecewise regression, primarily through formulations based on breakpoint indicator variables or segment assignment variables. Early work on segmented regression using integer programming can be traced back to [13] and [14], with more recent formulations incorporating modern MIP solvers for exact piecewise linear fitting [15], while a recent quadratic formulation based on big-M constraints can be found in [16].

In breakpoint-based formulations, binary variables indicate whether a changepoint occurs at a given time index, while continuous variables model segment-specific parameters. These models allow regression and segmentation decisions to be optimized jointly and can accommodate certain side constraints. However, constraints involving segment-level properties—such as length distributions, aggregate statistics of selected segments, or interactions between non-adjacent segments—typically require additional auxiliary variables and complex logical constraints, leading to weak linear relaxations and limited scalability. Assignment-based formulations, in which observations are assigned to segments via binary variables, suffer similar difficulties when enforcing contiguity and global structural constraints. In both cases, segmentation is encoded implicitly, rather than directly treating segments as decision objects.

The approach proposed in this paper differs fundamentally from the above paradigms by adopting a set-based perspective. Each feasible segment is explicitly represented as a decision variable, and the segmentation problem is formulated as a set covering problem in which the time axis must be covered once. While set covering and set partitioning models are classical tools in combinatorial optimization [17,18], they have received comparatively little attention in the changepoint detection literature.

By associating segments with decision variables, the set-covering formulation provides a natural mechanism for expressing global structural constraints directly at the segment level. Aggregate constraints on the number, length, distribution, or interaction of segments can be imposed through simple linear inequalities, without resorting to state augmentation or intricate logical constructs. This distinguishes the proposed SC-CPD framework from existing MIP formulations and enables a level of modeling expressiveness that is difficult to achieve within DP-based, shortest-path, or breakpoint-indicator models.

In summary, existing optimal CPD models exhibit a clear trade-off between computational efficiency and modeling flexibility. Dynamic programming and graph-based approaches provide excellent performance for additive objectives but offer limited support for global structural constraints. Prior MIP formulations increase modeling power but typically encode segmentation implicitly and encounter scalability issues when complex segment-level constraints are introduced.

The proposed SC-CPD framework occupies a complementary position in this landscape. By combining a set-covering formulation with modern MILP techniques and problem-specific preprocessing, it delivers global optimality while substantially expanding the class of constraints that can be handled in a principled and scalable manner. This positions SC-CPD as a flexible and extensible optimization framework for changepoint detection in applications where global structure and domain-specific requirements play a central role.

3. An extended set covering model

The model adopts a set partitioning perspective: all subsequences satisfying predefined structural properties—termed feasible runs—are generated and scored according to a segment-specific quality metric. The segmentation problem is then formulated as the selection of a collection of runs that forms a partition of the time index set while optimizing a global objective function based on residual loss or model fit. The resulting mixed-integer linear program provides a flexible framework in which additional modeling requirements can be seamlessly encoded via linear constraints and handled by state-of-the-art MIP solvers [19].

A Set Partitioning Problem (SPP) lies at the core of the formulation. The objective is to minimize an aggregate residual-based cost, while the constraints enforce that each time index of the series is covered exactly once by the selected segments. No restriction is imposed on the functional form of the segment model: it may consist of linear regression (yielding piecewise linear regression), but also of nonlinear or nonparametric specifications. Moreover, different runs may be associated with different model classes without altering the structure of the optimization problem.

Formally, let $X = [x_j]$, $j = 1, \ldots, nruns$, denote the binary decision variables, where $x_j = 1$ if run j is selected and $x_j = 0$ otherwise. Feasible runs may be generated so as to incorporate preliminary dominance rules, for instance by excluding segments that are too short or too long. Each run j is assigned a cost c_j, computed according to any suitable goodness-of-fit or loss metric proposed in the literature, including R^2, SER, χ^2, RMSE, variance-based criteria, etc. The covering constraints ensure that every series point is included in exactly one selected run.

The resulting Time Series Set Partitioning (TSSP) model is defined as follows. Let $a_{ij} \in \{0, 1\}$ be a coefficient equal to 1 if and only if run j covers time index i, for $i = 1, \ldots, npoints$ and $j = 1, \ldots, nruns$.

$$
\begin{aligned}
(TSSP) \quad z_{TSSC} = \min & \sum_{j=1}^{nruns} c_j x_j && && (1)\\
s.t. & \sum_{j=1}^{nruns} a_{ij} x_j \geq 1, && i = 1, \ldots, npoints && (2)\\
& \sum_{j=1}^{nruns} x_j = maxruns && && (3)\\
& x_j \in \{0, 1\}, && j = 1, \ldots, nruns && (4)
\end{aligned}
$$

Unfortunately, in real-world applications the number of feasible runs can be very large, making the resulting SPP computationally demanding and, in some cases, intractable within acceptable time limits. From a polyhedral perspective, the set partitioning formulation is typically tighter but also harder to explore computationally, as the equality constraints induce a more restrictive feasible region and a combinatorially demanding branching structure. To mitigate this issue, we relax the equality constraints 3 into inequalities, thereby transforming the formulation into a Set Covering Problem (SCP). The covering polyhedron is generally easier to handle algorithmically and can be solved in significantly larger dimensions, albeit with a weaker linear programming relaxation.

The relaxation allows multiple runs to cover the same series points, and therefore a postprocessing phase is required to extract a feasible segmentation from the (possibly overlapping) covering solution. The resulting set covering formulation is denoted by TSSC.

State-of-the-art MIP solvers are highly effective on SCP formulations; however, very large instances may still entail substantial computational effort. To further enhance scalability, [20] proposed a Lagrangian matheuristic [21] for the TSSC problem where all covering constraints (2) are relaxed by associating a Lagrangian multiplier λ_i with each constraint, $1 \leq i \leq npoints$, while retaining only the relaxed covering constraint in the formulation. The resulting Lagrangian model is solved via a subgradient optimization scheme. The corresponding Lagrangian subproblem is computationally simple: at each iteration, it amounts to selecting at most the *maxruns* variables with the most negative reduced costs. A simple fixing heuristic is embedded within the procedure: the incumbent solution of the subproblem, which may be infeasible with respect to the relaxed covering constraints, is iteratively augmented by adding selected variables until feasibility is restored. This relaxation makes it possible to easily include constraints such as the maximum number of segments allowed. However, in most cases, additional constraints must be added to the subproblem or relaxed themselves, which makes the approach impractical.

In any case, after solving a set covering relaxation, postprocessing is required to transform any SCP solution into a feasible SPP segmentation. This can be accomplished in linear time by scanning each region of overlapping runs and identifying the splitting point that minimizes the total cost obtained by assigning the preceding observations to the preceding segment and the subsequent observations to the following segment.

3.1. Cost function

Models TSSP and TSSC are suitable for offline (or anytime) analysis assuming additive costs. The model is independent on the specific function used to quantify the cost associated with each segment. The CPD literature proposes different cost functions, including negative log-likelihood (Horvath 1993; Chen and Gupta 2000), quadratic loss or cumulative sums (e.g., Inclan and Tiao 1994; Rigaill 2010), but any statistical measure of fit can be used for the task. Each measure favors different characteristics of the model and the results can be widely different.

In practical applications of changepoint detection (CPD), model quality is often judged less by its formal optimality properties and more by the interpretability of the resulting segmentation. In many real-world settings, the most useful segmentation is not necessarily the one that minimizes a given objective function, but rather the one that aligns with human perception of structural changes or provides a coherent narrative for subsequent decision-making. Accordingly, the utility of a cost function may depend as much on the clarity and explanatory power of its segmentation as on its predictive accuracy.

To reconcile algorithmic outputs with visual or domain-based expectations, many selection procedures incorporate penalty terms that regulate the number of detected changepoints. These penalizations effectively allow practitioners to adjust the segmentation to better match subjective expectations. However, the resulting models should be regarded as practically useful approximations rather than representations of an underlying "true" model, which is unlikely to be included among the finite set of candidate models considered in practice [10].

A further limitation of user-defined penalization schemes is that they implicitly require human feedback. In heterogeneous datasets, appropriate penalty calibration often needs to be tailored on a per-series basis, which limits scalability and automation. To reduce this dependency on manual tuning, we evaluated several alternative cost functions drawn from both information-theoretic and optimization-based frameworks, with the aim of identifying a more robust and resilient formulation suitable for automated CPD analysis.

We evaluated a broad range of cost functions, including AIC, BIC, Chi-square statistics, MSE, QRMSE, R^2, RMSE, SER, and simple variance; comparative visual results on a M6 benchmark instance are reported in Figure 1. No single criterion consistently dominates the others across all benchmark instances. Performance varies depending on the underlying signal structure, noise characteristics, and segment length distribution, supporting the claim that cost-function performance is inherently context-dependent.

For the subsequent analysis, we focus on two representative and complementary criteria: AIC and QRMSE. AIC is well established in the CPD literature, particularly through its equivalence to penalized negative log-likelihood formulations, and provides a theoretically grounded likelihood-based benchmark.

QRMSE (Quasi-RMSE) [20], by contrast, lies between MSE and RMSE in terms of segment-length sensitivity. Relative to MSE, it places less emphasis on short segments, while avoiding the stronger preference for long segments induced by RMSE. This intermediate behavior makes it attractive in applications where neither aggressive segmentation nor excessive smoothing is desirable.

Importantly, although QRMSE is additive across segments and can therefore be used within optimal partitioning frameworks, it does not satisfy the inequality condition required for pruning in algorithms such as PELT. In particular, the cost of a merged segment may be lower than the sum of the costs of two adjacent subsegments. This violation prevents the use of the pruning guarantees that underpin the computational efficiency of such methods, limiting the practical applicability of QRMSE in large-scale settings.

3.2. Global series constraints

The literature proposed many constraints that could be requested on the single segments of the model or on the whole modelled series. Typical constraints on single segments include

- Minimum spacing between changepoints: Enforce a global minimum gap (e.g., at least L points between any two changepoints) to prevent over-fragmentation of the series.
- Boundary exclusion: Disallow changepoints too close to the start or end (e.g., first/last K points), ensuring stable estimation at boundaries.
- Seasonal/calendar alignment: Restrict changepoints to a predefined set of admissible positions (e.g., month-ends, business-cycle markers), imposing global positional structure.
- Equal or bounded segment sizes: Require near-equal segment lengths or bound length variability (e.g., max/min segment length ratio), affecting the global partition shape.
- Group sparsity on positions: Impose structured sparsity over candidate positions (e.g., block or clustered positions), selecting changepoints from predefined groups globally.

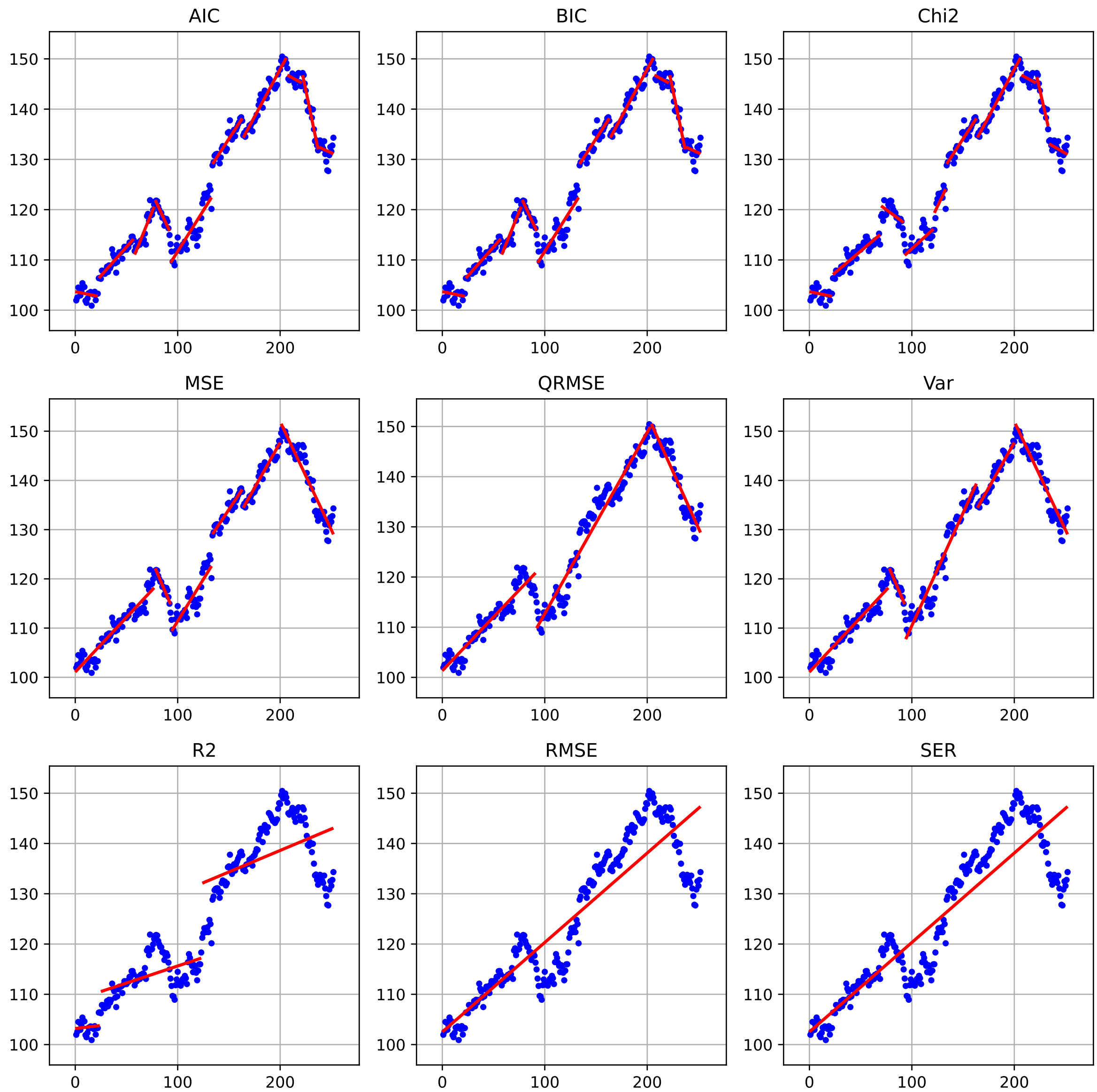


Figure 1. Alternative cost functions on series ROST of benchmark M6

- Template matching: Require changepoint positions to follow a given motif (e.g., known intervention schedule) with limited deviations across the whole timeline.

Many changepoint frameworks encode these ideas via global penalization rather than hard constraints—using penalties on the number of changepoints (*L0*), total variation (*L1*), or graph-structured transitions that enforce sign alternation or monotone regimes across the entire sequence. These approaches provide a way to control global behaviors, such as roughness or regime structure, and are usually implemented by dynamic programming or specialized algorithm limiting the possibility to flexibly expand the proposed model.

These are the constraints that "operate over the whole segmentation" and are discussed in the global constraint catalogues and CP-based changepoint detection literature (Beldiceanu, Régin; O'Hearn; Gionis; Killick; etc.), as well as in bioinformatics segmentation and shape-constrained segmentation. The main global constraints identified in the segmentation literature (CP + changepoint detection + structural break models) are:

- Segment-count constraints (you already use); In MIP: $\sum d_t \leq K$.
- Constraints on the distribution of segment lengths (global AMONG/NVALUES) Global-length constraint Constraint on the multiset of segment lengths, not per-segment one-by-one: (a) Total

length equality (trivial): $\sum length_i = T$ But more interesting: (b) Length profile constraints Examples seen in CP segmentation papers: AMONG/LENGTH-NVALUES constraints on the set of lengths e.g., number of distinct segment lengths leq L SUM of lengths in certain categories, e.g.: number of long segments $\leq \alpha$, number of short segments $\leq \beta$.

- Constraints on the global pattern of changes (REGULAR, PEAK/VALLY count) Used in shape-constrained segmentation, speech segmentation, and bioinformatics. A peak is an increase followed by a decrease in segment means (or vice versa). Global constraint: $PEAK_COUNT \leq P$.
- Global monotonicity or global shape constraints. Not per segment, but a global pattern constraint: Entire segmentation must be monotone nondecreasing, or Must follow a user-specified pattern (e.g., up $\rightarrow$ plateau $\rightarrow$ down). Define a finite automaton over segment types (increasing, decreasing, stationary). Segments must form a string in that language. Examples: No "up–down–up–down" oscillations beyond a pattern. Must alternate increase/decrease in a specific regular pattern. Change types must follow periodic structure (e.g., stable $\rightarrow$ volatile $\rightarrow$ stable).
- Global cost budgets / global total variation constraints. A constraint on the sum of segment costs, similar to a global knapsack or global linear constraint: Sum of all segment penalties leq B Sum of divergences leq B Total variation leq B (global TV constraint)
- Global constraint on number of distinct regimes. Constraint: Number of unique $regimes \leq R$. Used in regime-switching, hidden-state segmentation, and multiple-change models.
- Global maxima/minima on segment attributes (e.g., amplitude range). $Maxjump \leq \Delta_max$, $Minjump \leq \Delta_min$, Or bound the range of all segment means: $max(\mu_i)\check{} min(\mu_i) \leq R$.
- Global forbidden-pattern constraints. A global constraint that forbids entire patterns: No more than α alternations of direction. No pattern like "short–long–short" repeated. Prevent certain segment-type sequences entirely. This is essentially a global pattern-avoidance constraint
- Global window-based count constraints. If each segment has some attribute a_i (e.g., slope, variance, change point score), enforce: $\sum a_i \leq A$, or$A_min \leq \sum a_i \leq A_max$. Examples in segmentation: Global constraint on total curvature. Global constraint on total number of outliers explained by segmentation. Constraint on total misfit.
- Global constraints on total curvature or total slope
- Minimum / maximum number of "episodes" of a type For multi-type segmentation (e.g., up/down/flat segments): At most K episodes of type "up", regardless of length. At least M episodes of type "stable". This is global: it refers to the entire labeling.
- Global domain-specific constraints Found in applied changepoint papers in ecology, neuroscience, econometrics: At most one structural break per season / per macro-regime (the "season" or "macro-regime" spans multiple segments; still a global constraint) Segments must align with known block boundaries, but only globally (global periodicity constraints) At most one changepoint in the entire first half of the series (windowed-global count constraints)

Constraints for changepoint sets (MERGE WITH ABOVE)

- Minimum/maximum number of changepoints: ensure at least / at most a certain number to avoid under-segmentation in known nonstationary contexts.
- Total variation budget: Constrain the sum of absolute jump magnitudes across all changepoints to be below a threshold, controlling global roughness.
- Net drift constraint: Enforce that the signed sum of jumps equals a specified value (e.g., zero), ensuring the process starts/ends at comparable levels. Endpoint level constraints: Constrain the final level to lie within a target range relative to the initial level (e.g., bounded net change), coupling start/end globally.
- Maximum total jump magnitude: Cap the aggregate absolute change across all changepoints, distinct from the count of changepoints.
- Monotonic episode limits: Allow only a bounded number of monotone episodes (e.g., at most M runs of nondecreasing levels), acting globally over segments.

- Parity or count constraints: Enforce an even/odd number of changepoints, or fixed counts of specific jump types (e.g., exactly R upward jumps).
- Total penalty budget: Instead of a fixed penalty per changepoint, constrain the sum of penalties (data-adaptive or position-weights), globally bounding the objective's regularization.
- Symmetry constraints: Enforce symmetry of changepoint locations around the series midpoint, or mirrored jump magnitudes, for symmetric processes.
- Multi-scale consistency: Enforce that changepoints persist across coarse-to-fine aggregations (e.g., daily to weekly), selecting positions robust globally.
- Robustness-to-outliers budget: Limit the number of changepoints attributable to isolated spikes by bounding the cumulative outlier-attributed evidence globally.

1) Global pattern constraints (e.g., up–down constraints across all segments)

Reference

Hocking, T. D., Rigaill, G., Fearnhead, P., & Bourque, G. (2017). A Log-Linear Time Algorithm for Constrained Changepoint Detection. arXiv:1703.03352. — Constrained changepoint detection with affine constraints on adjacent segment means (e.g., up–down patterns). arXiv

How it maps This work imposes global constraints across the segmentation (not per segment) on the relative direction of adjacent changes and shows how to incorporate them in an optimal detection algorithm. It's widely cited for global directional constraints on changepoint sequences.

2) Global structural constraints / regular patterns under a global formulation (higher-level constrained segmentation)

Reference

Hocking, T. D., Rigaill, G., Fearnhead, P., & Bourque, G. (2018). Generalized Functional Pruning Optimal Partitioning (GFPOP) for Constrained Changepoint Detection in Genomic Data. arXiv:1810.00117. — Generalizes the pattern constraints to alternate up–down and other affine constraints across the whole segmentation. arXiv

How it maps The GFPOP work formalizes global constraints across segments as state transitions in a constrained model, allowing regular patterns across the entire solution rather than isolated local rules.

3) Penalized global cost / budget constraints on segmentation

While not always referred to as a global constraint in a CP sense, the core idea of a total budget / global penalty on the segmentation as a whole (affecting number of changepoints / model selection) is formalized in classic changepoint frameworks:

Reference (classical)

Killick, R., Fearnhead, P., & Eckley, I. A. (2012). Optimal Detection of Changepoints With a Linear Computational Cost. Journal of the American Statistical Association. — Introduces the PELT algorithm with a global penalty on number of changepoints. (You can also cite the Arxiv pdf preprint if needed.) (PMCID not available, but widely known in the literature.)

Related reference

Yao, Y. C. (1988). Estimating the Number of Change-Points via Schwarz' Criterion. Statistics & Probability Letters. — Framework for selecting number of changepoints via information criteria (global cost consideration).

These works treat the number of segments / overall penalty as part of the global optimization rather than a per-segment constraint.

4) Global structural break frameworks (econometrics)

Although econometric "structural break" models are not always couched as CP/MIP constraints, they formalize global constraints on the solution such as at most m structural breaks, with simultaneous estimation of break locations:

Reference (foundational)

Bai, J., & Perron, P. (1998). Estimating and Testing Linear Models With Multiple Structural Changes. Econometrica, 66(1), 47–78. — Provides methods for global estimation of multiple structural breaks with explicit control on the number of breaks. Boston University

Follow-up survey

Casini, A., & Perron, P. (2018). Structural Breaks in Time Series. Oxford Research Encyclopedia of Economics and Finance. — A comprehensive review including multiple break estimation and tests. Boston University

These capture global constraints on the entire segmentation such as the number and joint locations of breaks.

Constraint Programming Global Constraint References

These are general global constraints (REGULAR, NVALUES, AMONG, etc.) that could be adapted to enforce whole-solution properties in a segmentation model:

Reference (core text)

Rossi, F., van Beek, P., & Walsh, T. (Eds.). (2006). Handbook of Constraint Programming. Elsevier, ISBN: 978-0-444-52726-4. — Surveys global constraints (e.g., REGULAR for sequence patterns, global count constraints) across variables. cs.uwaterloo.ca

From the handbook you can specifically cite chapters on:

REGULAR constraint — constrains a sequence of decision variables to be accepted by a finite automaton (global pattern constraint).

NVALUES / AMONG / COUNT constraints — constrain global counts / distinct values of sets of variables.

(The Handbook entry lists these as global constraint types; you can cite the appropriate chapter when applying them to segmentation.)

Summary of Types and Representative References Constraint type Representative reference Notes Global up–down pattern / affine constraints Hocking et al. (2017) arXiv:1703.03352 Changepoint detection with global constraints on adjacent segments. arXiv

General global segmented pattern / regular structure Hocking et al. (2018) arXiv:1810.00117 GFPOP extends global pattern constraints over all segments. arXiv

Global cost / penalty budget Killick et al. (2012), Yao (1988) PELT penalized optimization, information criteria. Global selection of breaks Bai & Perron (1998) Econometrica Structural break global model constraints. Boston University

Global CP constraints (REGULAR, COUNT, etc.) Rossi et al. (2006) Handbook of Constraint Programming Catalog of global constraints that can enforce global structure. cs.uwaterloo.ca

3.3. Additional constraints

In many real-world applications, the analyst is not only interested in minimizing a fitting error, but also in enforcing interpretable and domain-driven properties on the resulting segmentation. The literature proposed many constraints that could be requested on the single segments of the model or on the whole modelled series.

3.3.1. Segment-level constraints

Segment-level constraints restrict the admissible properties of each segment independently of the rest of the segmentation.

Let a segment be defined by its start and end indices (s, e), with length $\ell_{s,e} = e - s + 1$, slope $\beta_{s,e}$ (in the case of linear regression), and fitting cost $c_{s,e}$. Typical segment-level constraints include:

- Minimum and maximum segment length. Bounds on the length of each segment prevent over-fragmentation and excessively long segments. In optimal partitioning approaches, minimum segment size constraints can be naturally incorporated. For instance, the PELT algorithm proposed in [7] allows restricting the set of admissible segmentations by enforcing a lower bound on segment length. This constraint is expressed as

$$\ell_{\min} \leq e - s + 1 \leq \ell_{\max}. \tag{5}$$

Minimum segment length constraints are commonly imposed in exact and approximate changepoint detection methods through parameters such as `min-size`.

- Minimum spacing between changepoints. Enforce a minimum gap of at least L points between consecutive changepoints. Some methods such as PELT [7] accomodate this type of restriction by limiting admissible segment endpoints according to a predefined minimum segment size. This constraint can be written as
$$e - s + 1 \geq L, \tag{6}$$
- Slope constraints. When segments are modeled via linear regression, constraints can be imposed on segment-specific parameters, such as the slope $\beta_{s,e}$. In particular, constrained changepoint detection under affine inequality restrictions between segment parameters has been studied in [22]. Typical examples include bounded slopes,
$$|\beta_{s,e}| \leq \beta_{\max}, \tag{7}$$
or directional constraints enforcing locally non-decreasing or non-increasing behavior.
- Boundary exclusion. Changepoints can be disallowed in the first and last K observations of the series to ensure stable parameter estimation. This can be enforced by restricting admissible start and end indices:
$$s \geq K + 1 \quad \text{and} \quad e \leq N - K, \tag{8}$$
where N denotes the length of the series.
- Admissible breakpoint positions. Changepoints may be restricted to a predefined set $\mathcal{A} \subseteq \{1, \ldots, N\}$ of admissible positions, such as calendar or seasonal markers. For example, in the Ruptures framework [3] , the jump parameter enforces that changepoints can only be selected on a subsampled grid of indices, effectively reducing the admissible set of breakpoint locations. In this case, segment boundaries must satisfy:
$$s, e \in \mathcal{A}. \tag{9}$$
These restrictions are useful both for computational efficiency and for incorporating prior structural information into the segmentation model.

3.3.2. Global series constraints

In constrast to segment-level constraints, global series constraints impose conditions that couple multiple segments and operate on the segmentation as a whole. Mixed-Integer Programming formulations naturally support this class of constraints through linear inequalities linking multiple segment-selection variables.

Let $d_t \in \{0, 1\}$ be the changepoint indicator at position t, and let K be the total number of segments.

Typical global constraints include:

- Global bound on the number of segments. A global bound on the number of selected segments controls the overall model complexity. This constraint can be written as
$$\sum_t d_t \leq K_{\max}. \tag{10}$$
Constrained dynamic programming algorithms under this setting are discussed in [22].

- Global budget on segmentation cost. The total cost of the segmentation can be bounded:
$$\sum_j c_j x_j \leq B, \tag{11}$$
where c_j denotes the cost of segment j. This formulation corresponds to the constrained counterpart of penalized segmentation model [7], obtained via Lagrangian duality.
- Global total variation. A bound on the aggregate magnitude of all changes across the series:
$$\sum_t |\mu_{t^+} - \mu_{t^-}| \leq C_{\max}, \tag{12}$$
where μ_{t^-} and μ_{t^+} denote the segment parameters before and after a changepoint. This constraint controls the overall amount of structural variation in the segmentation and limits the cumulative magnitude of successive jumps between adjacent segment parameters. It is closely related to total variation regularization and to the fused lasso formulation of [23], where an ℓ_1 penalty is applied to successive parameter differences.
- Global monotonicity or shape constraint. The entire sequence of segment parameters must satisfy a prescribed global ordering,
$$\mu_1 \leq \mu_2 \leq \cdots \leq \mu_K, \tag{13}$$
or follow a predefined shape such as up-down or down-up. These constraints impose a structural pattern on the whole segmentation. Monotonicity-constrained segmentation and related shape-restricted models are studied in [22].
- Global pattern constraints on change directions. Constraints on the sequence of increase/decrease signs across all changepoints, such as limiting the number of alternations or enforcing a regular pattern:
$$\text{ALT\_COUNT} \leq A_{\max}. \tag{14}$$
Pattern constraints on the directions of successive changes, including enforced up-down alternation (PeakSeg model) are explicitly formulated via dynamic programming in [22].
- Window-based global count constraints. Bounds on the number of changepoints within large predefined windows:
$$L_w \leq \sum_{t \in W} d_t \leq U_w, \tag{15}$$
where W is the time interval. These constraints regulate the distribution of changepoints over the entire series.

4. Computational experience

We implemented models TSSP and TSSC in C++ and ran them on a Windows 11 Intel i7 machine equipped with 32 Gb of RAM. The MIP solver used was CPLEX 22.11. All codes and data are available from the project repository.

4.1. Dynamic programming approaches

PELT, SN ecc.

4.2. Benchmarking and Comparative Performance

The computational results reported here, still to be considered as preliminary, are mainly obtained on environmental monitoring data series, which initially prompted this research. The series were produced in the context of the SMARTLAGOON project, an EU H2020 project, born with the primary objective of developing a tool to allow real-time monitoring, analysis, to predict socio-environmental evolution of the vulnerable area *Mar Menor*, which is the largest saltwater coastal lagoon in Europe.

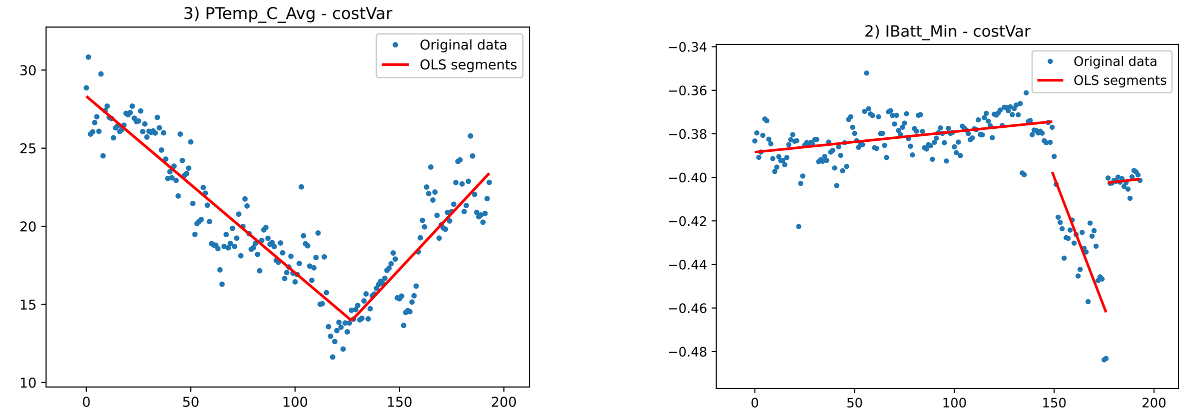
[24]. Within the framework of the project, smart buoys were deployed in the lagoon and their sensors generated series on 12 attributes.

The sensed variables used here are: the steam pressure (Vapor-Pressure-Avg), the average wind speed (WS-ms-Avg), water temperature measured by a thermistor at the depth of 0.5m (ThermTemp1-Avg), water temperature measured by oximeter at different depths (1m - Wtemp-C1-Avg, 3m - Wtemp-C2-Avg), water temperature measured by conductimeter at the depth of 1m (SDI-Temp-1m), minimum current in the battery (IBatt-Min), average and maximum temperature obtained by a datalogger panel temp thermocouple (PTemp-C-Avg, PTemp-C-Max) and average charging voltage (V-in-chg-Avg). The overall dataset was recorded from August 2022 to April 2023. We complemented these datasets with two others from different domains: economics (bitcoin-US dollar exchange rate, BTC-USD) and healthcare (COVID infections in Italy, Covid-Italia-22) to get a first indication of the generality of the approach across domains.

Table 1. Data series results

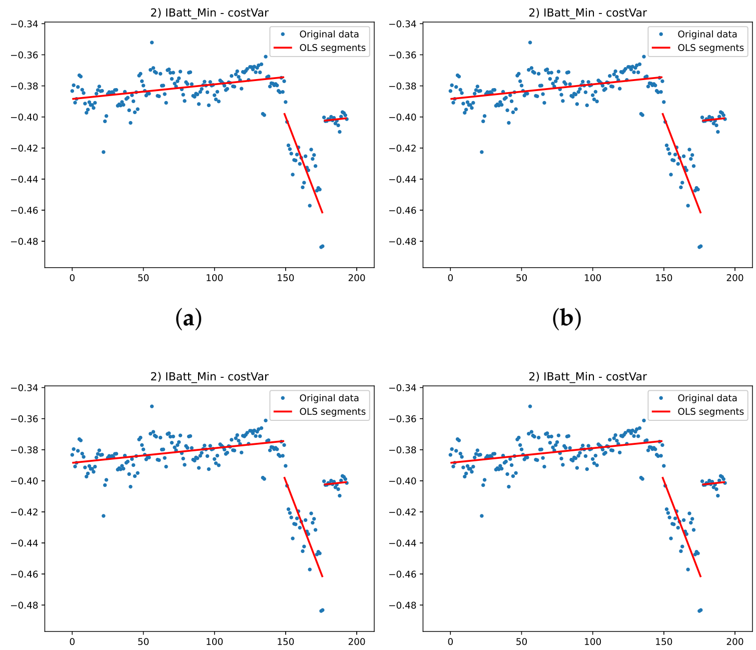
Dataseries	*npoints*	*nruns*	*nsegm*	*truns*	*tsolve*	*QMSE*	*QMSE1*	*QMSE2*
Vapor-Pressure-Avg	194	15400	6	0	1	0.90	1.14	1.03
Vapor-Pressure-Avg-2	1139	618828	7	30	832	2.85	3.26	3.32
WS-ms-Avg	1139	618828	2	30	1136	162	193.03	210.80
ThermTemp1-Avg	194	15400	8	0	1	5.12	5.98	na*
Wtemp-C1-Avg	1139	618828	19	30	1212	13.89	22.63	na*
Wtemp-C2-Avg	194	15400	8	0	1	50.75	68.15	na*
SDI-Temp-1m	1139	618828	21	30	1354	15.06	25.06	na*
IBatt-Min	194	15400	3	0	1	3.19e-3	5.29e-3	3.27e-3
PTemp-C-Avg	194	15400	5	0	1	46.40	68.56	58.12
PTemp-C-Max	194	15400	4	0	1	114.24	139.61	125.09
V-in-chg-Avg	194	15400	2	0	1	9.20	9.38	9.38
BTC-USD	366	60378	12	3	4	1.88e7	7.51e7	2.42e7
Covid-Italia-22	507	109746	5	-	11	5070.95	na	na

The results are summarized in table 1. The columns show the name of the series (*Dataseries*), the number of data points (*npoints*), the number of generated runs (*nruns*), the number of chosen segments (*nsegm*), the CPU time in seconds to generate all runs (*truns*), the CPU time in seconds to generate all runs (*truns*), the CPU time to solve the extended SCP model (*tsolve*), and a *quasi RMSE* quality measure of the solutions obtained by the algorithm described in section 3, *QMSE*, and, for comparison, the same measure for the solutions obtained by the codes of [25], *QMSE1*, and of [4], *QMSE2*. A comparison with the results from *Ruptures* [3] is being finalized and will be presented at the conference. Data with asterisks require special comments, which are incompatible with the page limit of this note, but will be presented at the conference and in the full version of the paper, *na*'s mean that no feasible solution was produced.



(**a**) Average temperature (**b**) Minimum battery current

Figure 2. Environmental dataseries

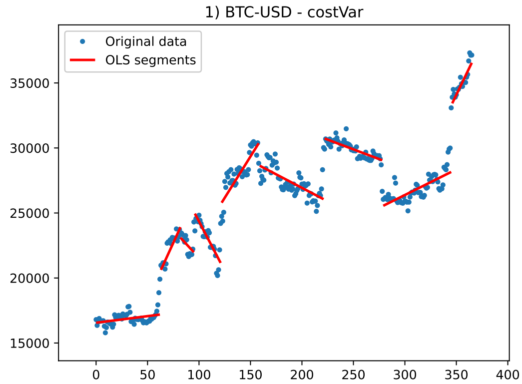


(**a**) (**b**) (**c**) (**d**)

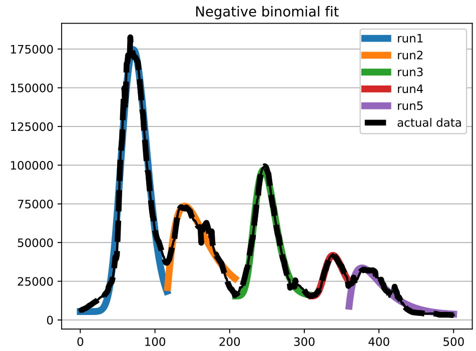
Figure 3. This is a wide figure. Schemes follow the same formatting. If there are multiple panels, they should be listed as: (**a**) Description of what is contained in the first panel. (**b**) Description of what is contained in the second panel. (**c**) Description of what is contained in the third panel. (**d**) Description of what is contained in the fourth panel. Figures should be placed in the main text near to the first time they are cited. A caption on a single line should be centered.

Not surprisingly, the set partitioning model could always produce the better solution, as it was the only algorithm that explicitly used the proposed cost function for optimization. More interesting is the limited computational time needed to solve instances with up to a few hundred variables, while the "curse of dimensionality" becomes apparent when solving instances with more than 1000 variables. The healthcare instance, here only a validation case, proves that the approach is also effective for nonlinear segmentation, but the search for optimal fitting parameters requires a very high CPU time. The figures 2 show the solution for two environmental data series, while the figures 4 show the two non environmental data series.

We emphasize that the validity of the proposed framework is not restricted to piecewise linear regression: any parametric or nonparametric trend function can be adopted for computing the segment-



(**a**) Bitcoin - USD exchange rate



(**b**) Covid Italy 2022

Figure 4. Economics and healthcare dataseries

specific error cost. Figure 5 illustrates the segmentation obtained by applying the model to the time series of total infected patients in Italy during the first year and a half (500 data points) of the Covid-19 pandemic. The epidemic evolved through successive waves, each of which can be effectively approximated by a trend following a negative binomial pattern. Accordingly, the formulation of Section 3 is applied by computing, for each feasible run, the fitting error associated with a negative binomial model. The subsequent optimization procedure remains unchanged.

As a final remark, the changepoints identified on this Covid-19 series admit a clear epidemiological interpretation, as they correspond to phases in which a new wave of contagion emerged or was reasonably expected to emerge.

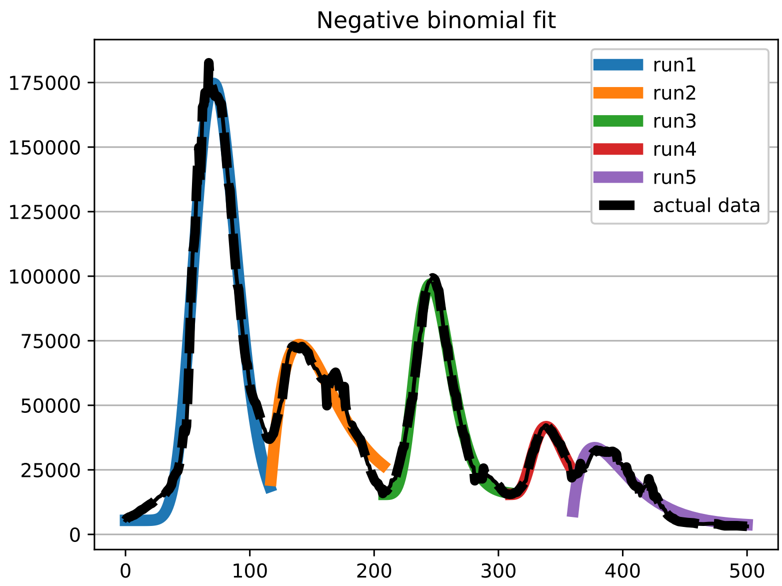


Figure 5. Covid diffusion in Italy

5. Conclusions

The paper presents an MIP-based approach, both a Lagrangian matheuristic and an exact model, to time series segmentation. The underlying mathematical model is able to adapt to different requirements on the resulting model, making it more tolerant to residuals or more representative of short trends. The analysis imposes no constraints on the model associated with each segment, which can be linear, polynomial, or defined on any distribution function of interest. It can even be a combination of different models, allowing for linearity on some segments and, for example, exponential increases on others.

This flexibility comes at the cost of having to solve a NP problem on large size instances. The increase in effectiveness of general MIP solvers allows to consider the solution of real-world instances to optimality, and it also provides the basis for the design of mathematically grounded heuristics. We report preliminary results on sensor data series obtained in an environmental monitoring, but also two experiments on financial and healthcare use cases.

The results so far provide only a first indication of the possibilities offered by mathematical models, but also of the difficulties to be overcome. Besides the obvious limit imposed by the NP-hardiness of the problem (the "curse of dimensionality"), which is only partially alleviated by heuristic solving, there are problem-specific issues to be faced, such as which cost function to use or which constraint to impose on the runs. In fact, the final result is only indirectly determined by our model, and there are still cases where results are unsatisfactory to the eye, even though they are numerically satisfying. Current results have already proven useful in the motivating application domain, both for missing value imputation and for short-term forecasting, obtained by decomposing the series down to the last segment and using the identified components to extend it into the near future. Moreover, these results do not seem to be affected by the application domain.

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Conflicts of Interest: "The authors declare no conflicts of interest."

Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
MIP	Mixed Integer Programming
NP	Nondeterministic Polynomial

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